

OpenClaw Plant Care Butler Tutorial

Before starting, please complete [01-Peripheral hardware configuration], [02-openclaw API-KEY configuration], [03-openclaw switching model], [04-OpenClaw prompt words], [05-AI voice interaction configuration], [06-Three dialogue scheme configuration tests] in the `Preparation before use` directory. If you are currently only using TUI/WhatsApp, you can skip 05 first. This article assumes that the basic environment has been prepared and only the plant care project itself will be launched.

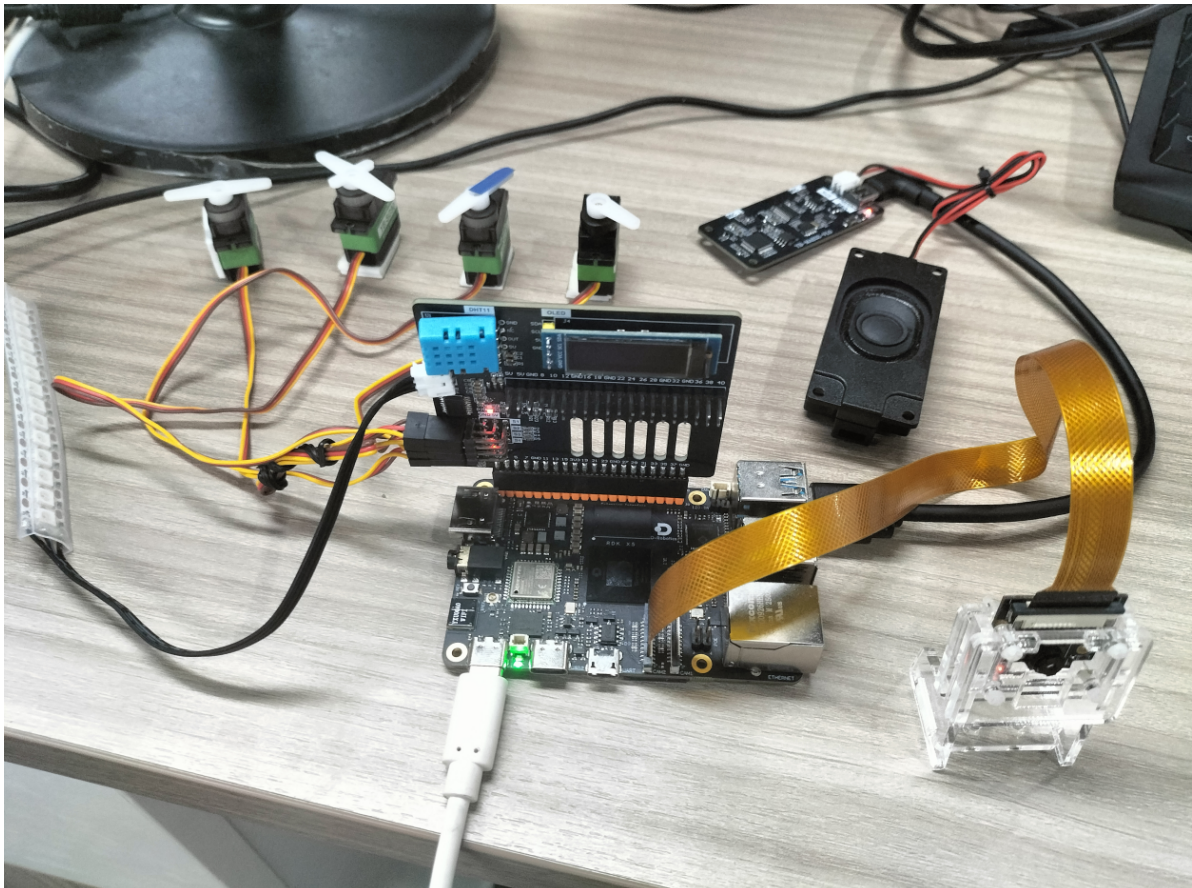
1. Tutorial Objectives

This article demonstrates OpenClaw's multi-peripheral collaboration capabilities through a plant care scenario.

- Learn how to combine temperature and humidity, OLED, RGB and cameras into a plant care assistant
- Converse with OpenClaw through WhatsApp, TUI, and voice code

2. Hardware preparation

- OpenClaw main control board
- DHT temperature and humidity sensor
- OLED display
- RGB light strip/RGB lamp bead module
- CSI camera
- [Optional] AI voice module + speaker
- Wiring diagram



3. Software preparation

Code entry that will be used in the tutorial

- Camera photo command: `/home/sunrise/hardware_hal_demo/camera`
- Visual analysis script: `/home/sunrise/.openclaw/workspace/skills/sunrise-vision/scripts/csi_describe`
- Voice entrance: `/home/sunrise/voice_to_openclaw/src/main.py`

This project essentially does not rely on a single "big project script", but lets OpenClaw orchestrate multiple basic tools:

- `dht` is responsible for reading the ambient temperature and humidity
- `oled` is responsible for displaying care information
- `rgb` is responsible for status reminders
- `camera` and `csi_describe` are responsible for observing the appearance of plants

4. Test the basic capabilities of the project

Before starting the current project, please complete a basic dialogue self-test of `openclaw tui`; if you have not done so yet, please read [06-Three dialogue scheme configuration tests] first. After passing, use TUI to verify the basic capabilities in the project.

Terminal input:

```
openclaw tui
```

```
sunrise@ubuntu:~/openclaw$ openclaw tui
🔌 OpenClaw 2026.3.24 (cff6dc9) - You had me at 'openclaw gateway start.'

openclaw tui - ws://127.0.0.1:18789 - agent main - session main
session agent:main:main

hello

你好呀，主人！🐾 我是小亚，随时待命。

有什么需要我帮忙的吗？
gateway connected | idle
agent main | session main (openclaw-tui) | deepseek/deepseek-v4-pro | tokens 12k/1.0m (1%)
```

After confirming that OpenClaw can communicate normally, you can try the following types of commands in sequence:

- Read current temperature and humidity
- Display "Plant OK" on OLED
- Change RGB lights to green
- Take a picture for me

If these individual capabilities can be run through, it means that the main links required for the project have been prepared.

5. Three options for talking to OpenClaw

Solution	Suitable scenario	Advantages	Suggested positioning
WhatsApp	Remote care, classroom demonstrations	Easy access, suitable for remote status viewing	Showcase "remote natural language care"
TUI	Local debugging, project joint debugging	The most direct and easiest to troubleshoot	Run through multiple peripheral links first
Voice code	Voice companionship, complete AI experience	Closer to the real assistant experience	As the final demonstration solution

If the AI voice module + speaker is optional, Feishu dialogue, tui, and WhatsApp dialogue need to broadcast the content of openclaw's reply. You can run the following program on the terminal. The AI voice module dialogue does not need to be run repeatedly

- Modify whether to open broadcast parameters

```
/home/sunrise/voice_to_openclaw/.env
```

```
ENABLE_TTS=true #This parameter is to control whether to broadcast. After modification, save it and run the following program on the terminal.
```

- Terminal input

```
cd ~/voice_to_openclaw
source .env/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) sunrise@ubuntu:~/voice_to_openclaw$ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/sunrise/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 WhatsApp solution

For WhatsApp access and robot creation steps, please refer directly to [06-Configuration Test of Three Dialogue Schemes]. This section only retains examples of conversations that would be appropriate for a plant care project.

After successfully creating the robot, you can remotely query the plant status in WhatsApp, for example:

- Take photos to see the current status of plants and perform plant care

20:10



< 2

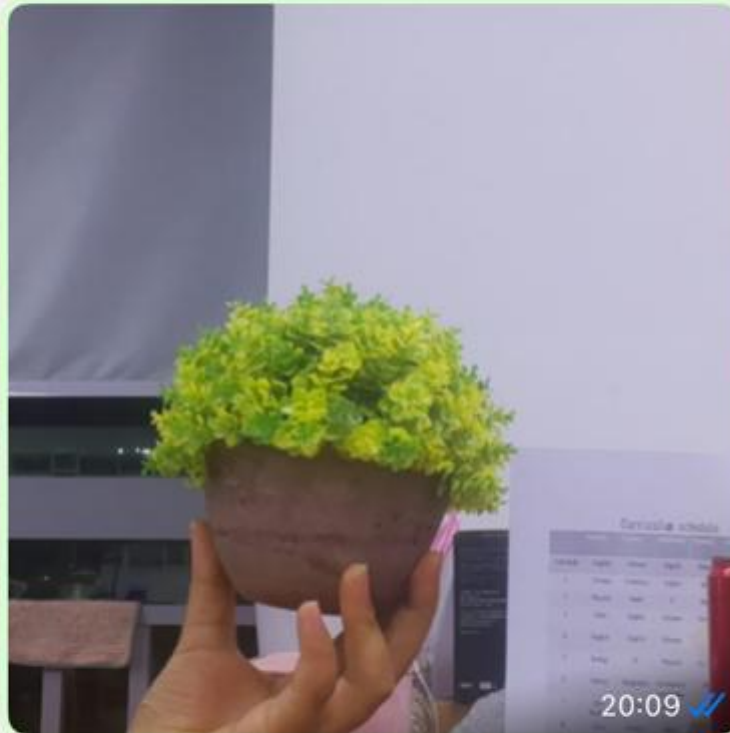


+86 153 3885 7526 (你)

给自己发消息

Take a photo to see the current plant status and perform plant care

20:09 ✓✓



I can see the plant! It's a small potted plant with vibrant green-yellow leaves. The foliage looks full and healthy. Let me complete the plant care workflow.

20:10 ✓✓

Plant care check completed!

Status:

- 📸 **Visual check:** Plant looks healthy with vibrant green-yellow foliage
- 🌡️ **Temperature:** 29.7°C (85.5°F)
- 💧 **Humidity:** 52%
- 📺 **OLED:** Displayed "Plant:"



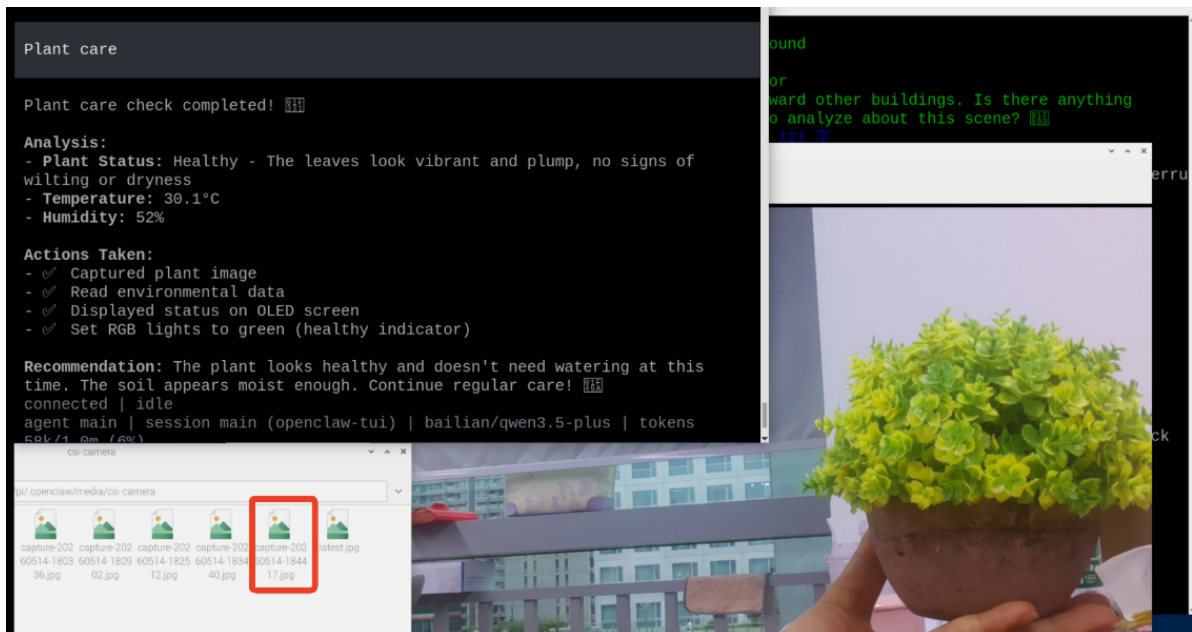
5.2 TUI scheme

If you have completed the TUI self-test in Part 6, enter the terminal text dialogue directly:

```
openclaw tui
```

Then you can directly enter project-related instructions, such as:

- Plant care



5.3 Voice code scheme

Requires optional ai voice interaction module + speaker

For voice module installation, environment configuration and basic wake-up process, please refer directly to [05-AI Voice Interaction Configuration]. After waking up, you can directly say the control content, such as:

- Help me check the status of the plants and whether it is suitable for watering now

```

[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音, 正在转写

[你] Me check the plant status and whether it's suitable for watering now.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Let me analyze the plant's condition and check the environment to
           determine if it needs watering.

## 🌿 Plant Status Check
           **Environmental Conditions:**
- **Temperature:** 30.1°C (86.2°F) -
           Quite warm! 🌡️
- **Humidity:** 52% - Moderate

**Plant Analysis:**
           Looking at the captured image, I notice:
- The plant has **bright
           green/yellow foliage** with uniform coloring
- Leaves appear
           **vibrant and upright** (no wilting visible)
- The foliage has an
           **artificial appearance** (plastic-like sheen)

**💧 Watering
           Recommendation:**

```

6. Project implementation

Note: The following are all dialogue demonstrations in `openclaw_tui` mode. You can also choose WhatsApp or voice mode to complete it.

6.1 Scenario 1: Environment reading

Experimental goals

Read the current temperature and humidity around the plants to complete the most basic environmental perception.

Effect description

OpenClaw will call DHT to read the current environment data and feedback the temperature and humidity in natural language.

Example instructions

- Read the current temperature and humidity near the plant
- What is the current ambient temperature and humidity?
- Help me check if the area around the potted plants is dry.

6.2 Scenario 2: Status Judgment

Experimental goals

Let OpenClaw give you a brief plant care recommendation after getting the environmental data.

Effect description

In addition to directly reporting values, OpenClaw can also summarize conclusions such as "dry", "hot" and "normal" in a way that is closer to daily language.

Example instructions

- Determine plant status based on current temperature and humidity
- Is it a good time to water?

6.3 Scene 3: Sound and light reminder

Experimental goals

When the environmental conditions are not ideal, use OLED and RGB to give more intuitive reminders.

Effect description

For example, it lights up green when the environment is normal, lights up yellow when it is hot, and lights up red when it is abnormal; at the same time, simple prompt information is displayed on the OLED, so that it is not just "speaking", but has clear peripheral feedback.

Example instructions

- If the environment is normal, the green light will turn on
- If it's too hot, turn on the yellow light and remind me
- Display "Plant Needs Care" on OLED

Expandable gameplay

- Linked buzzer for stronger reminder
- Made into a desktop care status light
- Add scheduled polling and automatic refresh

6.4 Scene 4: Taking photos and observing

Experimental goals

Use a camera to capture the current scene of the plant and let OpenClaw give a brief description.

Effect description

This link is suitable for observing the leaves, flower pot surface or overall growth. The focus of the current tutorial is "take a look and describe it" rather than automatically judging the disease, so it is more suitable for teaching demonstrations and lightweight auxiliary observation.

Example instructions

- Take a photo of the plant
- See how the leaves are doing
- Describe the current plant scene after taking the photo

